

Hariton Pitsik

ML ENGINEER · AI RESEARCH · DATA SCIENTIST

Saratov, Russia

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Education

Saratov State University

BACHELOR | FUNDAMENTAL COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Saratov, Russia

September 2023 - June 2027

Skills

Languages

Python, Rust, SQL

Communication languages

Russian (native), English (B1-B2)

Technologies

Git & GitHub, Docker, Linux, LaTeX, Typst, Jupyter, NumPy, Pandas, Matplotlib & Seaborn, Scikit-learn, XGBoost, CatBoost, FastAPI, ClearML, Streamlit, PyTorch, Transformers, PostgreSQL

Concepts

Linear Algebra, Mathematical Analysis, Probability Theory & Statistics, Computer Science, Classic ML Algorithms (Supervised & Unsupervised), Deep Learning, Natural Language Processing (NLP), Large Language Models (LLM)

Achievements

Teaching

- I believe that education is awesome and that we need to talk about it. I teach a machine learning club as part of a students community at the university.
- Recordings of the lectures on NumPy ([YouTube](#)) and Pandas ([YouTube](#)).
- Playlist of machine learning club: ([YouTube](#) [Here](#))

Competitions

- Participant of HSE & X5 contest “Growth gradient”, which focused on solving time series forecasting problems ([Here](#));
- As an analyst of existing hashing methods, the FSP hackathon participant examined potential sources of chaos.

Other activity

Projects

Local RAG System

 github.com/haritonn/pdf_rag

- Built a local RAG system with switchable LLM backend;
- Implemented hybrid retrieval, document chunking and automated relevance evaluation;
- Qdrant vector database and Streamlit interface application

Transformers from Scratch

 github.com/haritonn/transformers-scratch

- Implemented the full encoder-decoder architecture in PyTorch;
- Built a modular training pipeline for seq2seq tasks;
- Fully customizable pipeline for experiments.

Caption Generator

 github.com/haritonn/caption_gen

- Built an image captioning pipeline with ResNet backbone and an attention-based LSTM;
- Project wrote using only PyTorch, without pre-built implementations;

- Beam and greedy search decoding;
- End-to-end training workflow.

ResNet with Attention

 github.com/haritonn/resnet_attention

- Implemented baseline ResNet and attention-augmented ResNet variants in PyTorch;
- Compared convergence speed and classification accuracy;
- Tracked experiments with ClearML.

Courseworks

2nd Year: Generative Computer Vision

 github.com/haritonn/coursework2

COMPARED DIFFUSION MODELS, GANS, AND LMM-BASED APPROACHES FOR VIRTUAL TRY-ON. EVALUATED VISUAL QUALITY AND INFERENCE TIME ON A CUSTOM DATASET.

3rd Year: Analysis of Geometric Properties of Embeddings

ONGOING COURSEWORK FOCUSED ON THE ANALYSIS OF EMBEDDING SPACES.